

Nick's Question 0.1 Details from Remark 1.1 page 3 that part (i) implies the existence of K .

Is this correct in showing

Let $\psi \in C_0^\infty(\mathbb{R}^d)$ then we have

$$\begin{aligned} \int_{\mathbb{R}^d} dx \gamma(x) \psi(x) &= \int_{\mathbb{R}^d} dx \varphi(x) \mathcal{F}(\psi)(x) \\ &= \int_{\mathbb{R}^d} dx \mathcal{F}(\varphi)(x) \psi(x) \\ &= \int_{\mathbb{R}^d} dx \mathcal{F}(\sqrt{\varphi} \sqrt{\varphi})(x) \psi(x) \\ &= \int_{\mathbb{R}^d} dx \left(\mathcal{F}(\sqrt{\varphi}) * \mathcal{F}(\sqrt{\varphi}) \right)(x) \psi(x) \end{aligned}$$

and since this is true for all test functions ψ then we must have $\gamma(x) = \left(\mathcal{F}(\sqrt{\varphi}) * \mathcal{F}(\sqrt{\varphi}) \right)(x)$

Nick's Question 0.2 top of page 6

Why does assumption A imply that $\alpha \in (0, 2 \wedge d)$?

Nick's Question 0.3 page 8

Why do we have the representation (2.3)?

Nick's Question 0.4 Lemma 2.3

1. Why does $f(cA) = c^{d-\alpha} f(A)$ imply that when f is a density then $f(cx) = c^{-\alpha} f(x)$?
2. Does

$$\int_{\mathbb{R}^d} |\xi|^2 |\mathcal{F}g(\xi)|^2 d\xi = \int_{\mathbb{R}^d} |\nabla g(x)|^2 dx?$$

3. Why does $(\tilde{g}^2 * f)(x) = B^{-\alpha} (g^2 * f)(B^{-1}x)$. Here is what I have

$$\begin{aligned} B^{-\alpha} (g^2 * f)(B^{-1}x) &= B^{-\alpha} \int_{\mathbb{R}^d} g^2(y) f\left(\frac{x}{B} - y\right) dy \\ &= \int_{\mathbb{R}^d} g^2(y) f(x - By) dy \end{aligned}$$

since $B^{-\alpha} f\left(\frac{x}{B} - y\right) = f(x - By)$. Then,

$$\begin{aligned} \int_{\mathbb{R}^d} g^2(y) f(x - By) dy &= B^{-d} \int_{\mathbb{R}^d} g^2\left(\frac{u}{B}\right) f(x - u) du \\ &= B^{-d/2} \int_{\mathbb{R}^d} \tilde{g}^2(u) f(x - u) du \\ &= B^{-d/2} (\tilde{g}^2 * f)(x) \end{aligned}$$

and this shows that

$$B^{-d/2} (\tilde{g}^2 * f)(x) = B^{-\alpha} (g^2 * f)(B^{-1}x)$$

4. If I am correct on (3) then

$$\langle \tilde{g}^2 * f, \tilde{g}^2 \rangle = B^{d-\alpha} \langle g^2 * f, g^2 \rangle$$

5. If I am incorrect in (3) then how do we use $(\tilde{g}^2 * f)(x) = B^{-\alpha}(g^2 * f)(B^{-1}x)$ to get that $\langle \tilde{g}^2 * f, \tilde{g}^2 \rangle = B^{-\alpha} \langle g^2 * f, g^2 \rangle$. We have that

$$\begin{aligned} \langle \tilde{g}^2 * f, \tilde{g}^2 \rangle &= \int_{\mathbb{R}^d} (\tilde{g}^2 * f)(x) \tilde{g}^2(x) dx \\ &= \int_{\mathbb{R}^d} B^{-\alpha}(g^2 * f)(B^{-1}x) \tilde{g}^2(x) dx \\ &= \int_{\mathbb{R}^d} B^{-\alpha}(g^2 * f)(B^{-1}x) B^{-d/2} g^2(B^{-1}x) dx \\ &= B^{-\alpha-\frac{d}{2}} \int_{\mathbb{R}^d} (g^2 * f)(B^{-1}x) g^2(B^{-1}x) dx \\ &= B^{-\alpha+\frac{d}{2}} \int_{\mathbb{R}^d} (g^2 * f)(u) g^2(u) du \\ &= B^{-\alpha+\frac{d}{2}} \langle g^2 * f, g^2 \rangle \end{aligned}$$

and this is also different from the paper.

Nick's Question 0.5 page 11 When $d = 3$ we know that $G(t, \cdot)$ is a measure so does $(G(t, \cdot) * p_\epsilon)(x) = \int p_\epsilon(x - y) G(t, dy)$?

Nick's Question 0.6 page 11 typos

1. on the middle of page you say $d \leq 3$ right before (3.6) but I think it need to be $d = 3$.
2. When defining L and L_ϵ . There is a dx in both integrals but there needs to be a dt .
3. When applying the change of variables to L_n and $L_{n,\epsilon}$ the index on the sum I think should be Σ_n not S_n .

Nick's Question 0.7 How are the equations on the bottom of page 11 proved. For example, using (3.3), we see that

$$L_1(y_1) = \int_0^\infty e^{-t} \int_0^t G(s, y_1) ds dt$$

but according to the equation for L_n on the bottom of page 11 we should also have

$$L_1(y_1) = L(y_1)$$

where

$$L(x) = \int_0^\infty e^{-t} G(t, x) dt$$

But it doesnt seem that $L_1(y_1) = L(y_1)$. For example, consider $d = 1$ then we have that

$G(t, x) = \frac{1}{2} 1_{|x| < t}$ and so

$$\begin{aligned}
L_1(y_1) &= \int_0^\infty e^{-t} \int_0^t G(s, y_1) ds dt \\
&= \frac{1}{2} \int_0^\infty e^{-t} \int_0^t 1_{\{|y_1| < s\}} ds dt \\
&= \frac{1}{2} \int_0^\infty e^{-t} \int_{|y_1| \wedge t}^t ds dt \\
&= \frac{1}{2} \int_0^\infty e^{-t} (t - (|y_1| \wedge t)) dt \\
&= \frac{1}{2} \int_{|y_1|}^\infty e^{-t} (t - |y_1|) dt
\end{aligned}$$

However

$$\begin{aligned}
L(y_1) &= \int_0^\infty e^{-t} G(t, y_1) dt \\
&= \frac{1}{2} \int_0^\infty e^{-t} 1_{\{|y_1| < t\}} dt \\
&= \frac{1}{2} \int_{|y_1|}^\infty e^{-t} dt
\end{aligned}$$

and these do not seem to be equal.

Nick's Question 0.8 Lemma 3.3 What are the details showing

$$\int_{(\mathbb{R}^d)^n} \left[\int_{(\mathbb{R}^d)^n} \prod_{k=1}^n K(x_k - y_k) L_n(y_1, \dots, y_n) dy \right]^2 dx = \int_{(\mathbb{R}^d)^n} |\mathcal{F} L_n(\xi_1, \dots, \xi_n)|^2 \mu(d\xi_1) \dots \mu(d\xi_n)$$

for example suppose that $n = 1$ so we want to show

$$\int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} K(x - y) L_n(y) dy \right]^2 dx = \int_{\mathbb{R}^d} |\mathcal{F} L_n(\xi)|^2 \mu(d\xi)$$

By Plancherels Theorem we have

$$\int_{\mathbb{R}^d} \left[\int_{\mathbb{R}^d} K(x - y) L_n(y) dy \right]^2 dx = \int_{\mathbb{R}^d} \left[\mathcal{F} \left(\int_{\mathbb{R}^d} K(\cdot - y) L_n(y) dy \right) (\xi) \right]^2 d\xi$$

However anyway I continued seemed not to work. I thought we should use the property that $\mathcal{F}(f)\mathcal{F}(g) = \mathcal{F}(f * g)$ and this would give us

$$\int_{\mathbb{R}^d} \left[\mathcal{F} \left(\int_{\mathbb{R}^d} K(\cdot - y) L_n(y) dy \right) (\xi) \right]^2 d\xi = \int_{\mathbb{R}^d} \mathcal{F} \left(\int_{\mathbb{R}^d} K(\cdot - y) L_n(y) dy * \int_{\mathbb{R}^d} K(\cdot - y) L_n(y) dy \right) (\xi) d\xi$$

but continuing from this didnt seem to work.

Nick's Question 0.9 Lemma 3.4

1. Why does $\gamma(y_k - y'_k) = \int_{\mathbb{R}^d} K(y_k - x_k)K(y'_k - x_k)dx_k$. We know that $\gamma = K * K$ so

$$\begin{aligned}\gamma(y_k - y'_k) &= \int_{\mathbb{R}^d} K(y_k - y'_k - x_k)K(x_k)dx_k \\ &= \int_{\mathbb{R}^d} K(y_k - u)K(u - x_k)du\end{aligned}$$

by letting $u = y'_k + x_k$

2. Where are the first and second equalities coming from?

Nick's Question 0.10 Lemma 3.6 page 14

How do we get the inequality right above equation (3.13)?

Nick's Question 0.11 page 14

How does (1.22) imply the scaling property (3.15)?

Nick's Question 0.12 Proof of Theorem 3.1

In the first line of the proof of Theorem 3.1 it says that by theorem 2.2 it suffices to show that

$$\sum_{n \geq 0} \theta^n n! \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 < \infty$$

but this is not quite true since Theorem 2.2 says we need

$$\sum_{n \geq 0} \theta^n n! \left\| \tilde{f}_n(\cdot, x, t) \right\|_{\mathcal{H}^{\otimes n}}^2 < \infty$$

for every $x \in \mathbb{R}^d$.

Nick's Question 0.13 Proof of Theorem 4.1

1. I am confused about the first line of the proof? Why does

$$\mathbb{E}(|u(t, x)|^2) = \sum_{n \geq 0} \theta^n n! \left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2$$

According to theorem 2.2 it should be

$$\mathbb{E}(|u(t, x)|^2) = \sum_{n \geq 0} \theta^n n! \left\| \tilde{f}_n(\cdot, x, t) \right\|_{\mathcal{H}^{\otimes n}}^2$$

but then the rest of this proof would be incorrect since the very next line you use the scaling property for $\left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2$.

2. In part (iii) there is a scaling property used for $\left\| \tilde{f}_n(\cdot, x, t) \right\|_{\mathcal{H}^{\otimes n}}^2$ but a scaling property has never been proved for this, only $\left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2$. You said that by the scaling

property that

$$\mathbb{E}(|u(t, x)|^2) = \sum_{n \geq 0} \theta^n n! \left\| \tilde{f}_n(\cdot, x, t) \right\|_{\mathcal{H}^{\otimes n}}^2 = \sum_{n \geq 0} \theta^n n! t^n \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2$$

but this doesn't seem correct since if this is supposed to be true for every $x \in \mathbb{R}^d$ then this would mean that

$$\left\| \tilde{f}_n(\cdot, 0, t) \right\|_{\mathcal{H}^{\otimes n}}^2 = t^n \left\| \tilde{f}_n(\cdot, 0, 1) \right\|_{\mathcal{H}^{\otimes n}}^2$$

which is not the same as the scaling property (3.15).

Nick's Question 0.14 Lemma 6.2 proof After doing the substitution, we have

$$\begin{aligned} & \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \phi(\xi_k) \int_0^\infty e^{-t} \int_{[0, t]^n <} \prod_{k=1}^n \mathcal{F}G(s_k - s_{k-1})(\xi_k, \dots, \xi_n) ds dt \mu(\xi_1) \dots \mu(\xi_n) \\ &= \int_{(\mathbb{R}^d)^n} \prod_{k=1}^n \phi(\xi_k) \left(\int_0^\infty e^{-u} du \right) \prod_{k=1}^n \left(\int_0^\infty e^{-u_k} \mathcal{F}G(u_k, \cdot)(\xi_k, \dots, \xi_n) du_k \right) \mu(\xi_1) \dots \mu(\xi_n) \end{aligned}$$

Where does the e^{-u_k} come from? Since we let $t - s_n = u$ shouldn't there only be a e^{-u_n} ?

Also I see how $u_i \in (-\infty, \infty)$ shows that $\mathbf{s} \in [0, t]^n <$ but how could we see that $\mathbf{s} \in [0, t]^n <$ shows that $u_i \in (-\infty, \infty)$?

Nick's Question 0.15 Prop 6.3 proof At the equality above (6.6), why were we able to remove the $\frac{1}{n} \sum_{\sigma \in \Sigma_n}$?

Nick's Question 0.16 In (6.9) where does the $\sum_{\sigma \in \Sigma_n}$ come from?

Nick's Question 0.17 page 29 Three centered lines starting with $\rho(\mathcal{F}f)$

Nick's Question 0.18 I might have something that shows that

$$H(\xi) := \int_{\mathbb{R}^d} \frac{h(\xi + \eta)h(\eta)}{\sqrt{1 + |\xi + \eta|^2} \sqrt{1 + |\eta|^2}} d\eta$$

is even if and only if h is even.

First suppose that h is even. Then,

$$\begin{aligned} H(-\xi) &= \int_{\mathbb{R}^d} \frac{h(-\xi + \eta)h(\eta)}{\sqrt{1 + |-\xi + \eta|^2} \sqrt{1 + |\eta|^2}} d\eta \\ &= \int_{\mathbb{R}^d} \frac{h(\xi - \eta)h(\eta)}{\sqrt{1 + |\xi - \eta|^2} \sqrt{1 + |\eta|^2}} d\eta \\ &= \int_{\mathbb{R}^d} \frac{h(\xi + u)h(-u)}{\sqrt{1 + |\xi + u|^2} \sqrt{1 + |-u|^2}} du \end{aligned}$$

$$\begin{aligned}
&= \int_{\mathbb{R}^d} \frac{h(\xi + u)h(u)}{\sqrt{1 + |\xi + u|^2}\sqrt{1 + |u|^2}} du \\
&= H(\xi)
\end{aligned}$$

and so H is even.

Now suppose that h is not even. Then,

$$\begin{aligned}
H(-\xi) &= \int_{\mathbb{R}^d} \frac{h(-\xi + \eta)h(\eta)}{\sqrt{1 + |-\xi + \eta|^2}\sqrt{1 + |\eta|^2}} d\eta \\
&= - \int_{\mathbb{R}^d} \frac{h(\xi + u)h(u + 2\xi)}{\sqrt{1 + |\xi + u|^2}\sqrt{1 + |u + 2\xi|^2}} d\eta
\end{aligned} \tag{0.1}$$

cont1

This is incorrect, there should not be a minus sign here. This proof is incorrect. where above we use the substitution $-\xi + \eta = \xi + u$. Recall that our problem deals with looking at

$$\sup_{\|h\|_{L^2(\mathbb{R}^d)}=1} \int_{\mathbb{R}^d} \mathcal{F}f(\xi) \left[\int_{\mathbb{R}^d} \frac{h(\xi + \eta)h(\eta)}{\sqrt{1 + |\xi + \eta|^2}\sqrt{1 + |\eta|^2}} d\eta \right] d\xi$$

where $\mathcal{F}f(\cdot)$ is non-negative and non-negative definite. Since it is non-negative we can restrict our choices of h to also be non-negative and in particular we have that

$$\int_{\mathbb{R}^d} \frac{h(\xi + u)h(u + 2\xi)}{\sqrt{1 + |\xi + u|^2}\sqrt{1 + |u + 2\xi|^2}} d\eta \geq 0$$

but then this means that

$$- \int_{\mathbb{R}^d} \frac{h(\xi + u)h(u + 2\xi)}{\sqrt{1 + |\xi + u|^2}\sqrt{1 + |u + 2\xi|^2}} d\eta \leq 0. \tag{0.2}$$

cont2

But (0.1) and (0.2) then implies that

$$H(-\xi) \leq 0$$

and since we assume $h \geq 0$ then $H(\xi) \geq 0$ for all $\xi \in \mathbb{R}^d$ and this mean $H(\xi) = 0$ which can not always be the case. Thus we must have h to be even in order to get a non-zero $H(\cdot)$.